

FASTAPI QUESTIONS AND ANSWERS

1 . What is CRUD in FastAPI?

* It stands for Create, Read, Update, and Delete. These are the fundamental actions performed on a database through an API to manage information.

2. Why do we use the POST method for inserting data?

POST is designed to send data to the server to create a new resource. Unlike GET, it carries data in the request body, which allows for larger payloads and better security for sensitive information.

3. Difference between PUT and PATCH

Feature	put	patch
Purpose	Replaces the entire resource	Updates specific parts of the resource.
Data Required	You must send the full object (all fields)	You only send the fields you want to change.
Missing Fields	FieldsIf a field is missing, it is often set to null or default.	Missing fields remain unchanged in the database.

4. What is the path parameter?

A path parameter is a variable part of the URL (**e.g.**, `/items/{item_id}`) used to identify a specific resource. It is mandatory and embedded directly in the endpoint path.

5. What is the request body in FastAPI?

The request body is the data sent by the client to the API, typically in JSON format. It is used when you need to send a complex data object rather than just simple parameters in the URL.

6. Why do we use Pydantic models?

Pydantic models are used for data validation and type hinting. They ensure the incoming request body matches the expected format, automatically converting types and returning clear error messages if the data is invalid.

7. What status code is returned for successful deletion?

Typically, 204 No Content is used if there is no body to return, or 200 OK if a confirmation message is included.

8. How can we check whether a student exists before updating?

You can perform a database query using the student's ID at the start of your update function. If no record is found, you should raise an

`HTTPException(status_code=404, detail="Student not found")` to stop the process.

9. What happens if duplicate IDs are inserted?

If the ID is a Primary Key or has a Unique Constraint in the database, the insertion will fail and raise a database error (e.g., `IntegrityError`), which the API should catch and return as a `400 Bad Request` or `409 Conflict`.

10. How does FastAPI automatically generate Swagger UI?

FastAPI generates documentation using OpenAPI.

- Answer: Auto-documentation
- Explanation: FastAPI reads your code's type hints and path operations to build a standard OpenAPI schema, which is then rendered visually by Swagger UI at the `/docs` endpoint.